

Dr. Mrityunjoy Saha

Ph.D. (Fluid Mechanics) | Director's Fellow
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EDUCATION

- Ph.D. in Applied Mathematics (Fluid Mechanics)** 2021 – 2026
Indian Institute of Technology Jodhpur
Thesis: Hydrodynamic Stability Analysis on a Class of Complex Flow Systems
CGPA: 8.69/10
- M.Sc. in Mathematics (Applied Mathematics)** 2018 – 2020
Cooch Behar Panchanan Barma University
CGPA: 8.19/10
- B.Sc. (Hons.) in Mathematics** 2015 – 2018
Dinhata College, Cooch Behar Panchanan Barma University
Percentage: 62.5

TECHNICAL SKILLS

- Programming & Computing**
Python, C/C++, MATLAB, Mathematica
- Scientific Computing & Simulation**
COMSOL Multiphysics, High-Performance Computing (HPC)
- Technical Expertise**
Numerical Methods, CFD, Stability Analysis, Spectral Methods, Numerical PDEs

RESEARCH EXPERIENCE

- Director's Fellow, IIT Jodhpur** January 2026 – Present
Developing high-order numerical solvers and stability frameworks for time-dependent complex flow systems.
- Senior Research Fellow, IIT Jodhpur** 2024 – 2025
Developed spectral collocation-based numerical solvers for stability analysis of MHD-induced anisotropic porous channel flows and flows in oscillating bounded or semi-bounded domains.
- Junior Research Fellow, IIT Jodhpur** 2022 – 2023
Developed mathematical models to analyze rotational instability mechanisms in wall-bounded porous media flows.

PUBLICATIONS

- M. Saha** and S. Ghosh – [Physics of Fluids](#) (2026).
- M. M. Hossain, **M. Saha**, S. Ghosh, and H. Behera – [Physics of Fluids](#) (2025).
- M. Saha**, S. Sengupta, S. Mukhopadhyay, and S. Ghosh – [Computers & Fluids](#) (2025).

ONGOING RESEARCH

- Modal and non-modal instability of a pressure-driven flow through an anisotropic porous material: Brinkman-Forchheimer model. (Under review)
- Dual instability in a vertical crossflow-induced oscillating channel flow. (Submitted)
- Non-modal instabilities in Poiseuille flow in inhomogeneous porous media. (Ready for submission)
- Effect of uniform vertical crossflow on rotational instability in a channel flow. (In preparation)

TECHNICAL TRAINING & WORKSHOPS

- **Python Programming for Computational Fluid Dynamics Applications**
Electronics & ICT Academy, IIT Guwahati (June-July, 2024)
- **Introduction to Computational Fluid Dynamics with Practical Applications**
Department of Mechanical Engineering, IIT Indore (July, 2022)

CONFERENCE PRESENTATIONS

- **78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2025)**
George R. Brown Convention Center, Houston, Texas, USA.
- **International Conference on Recent Advances in Fluid Mechanics and Nanoelectronics 2023**
Manipal Institute of Technology, Bengaluru, India.

TEACHING EXPERIENCE

- **Teaching Assistant** **2022 – 2026**
Indian Institute of Technology Jodhpur
Courses assisted: Scientific Computation, Mathematical Modeling, Data Science, Linear Algebra, Applied Mathematics & Statistics.

AWARDS AND SCHOLARSHIPS

- Director's Fellowship, Indian Institute of Technology Jodhpur.
- American Physical Society - Division of Fluid Dynamics Travel Award 2025.
- ANRF International Travel Support for APS DFD 2025.
- CSIR Foreign Travel Grant for APS DFD 2025.
- Qualified Graduate Aptitude Test in Engineering (GATE), Mathematics (2022, 2023).
- DST INSPIRE Scholarship - Government of India.
- SVMCM Scholarship - Government of W.B. (UG, PG, and Higher Secondary).

REFERENCE

- **Dr. Sukhendu Ghosh** (Ph.D. Supervisor)
Associate Professor, Department of Mathematics, IIT Jodhpur
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